CasA: A Cascade Attention Network for 3-D Object Detection From LiDAR Point Clouds

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***Abstract*— Three-dimensional object detection from light detection and ranging (LiDAR) point clouds has gained great attention in recent years due to its wide applications in smart cities and autonomous driving. Cascade framework shows its advancement in 2-D object detection but is less investigated in 3-D space. Conventional cascade structures use multiple *separate* subnetworks to sequentially refine region proposals. Such methods, however, have limited ability to measure proposal quality in all the stages, and it is hard to achieve a desirable performance improvement in 3-D space. This article proposes a new cascade framework, termed Cascade Attention (CasA), for 3-D object detection from LiDAR point clouds. CasA consists of a region proposal network (RPN) and a cascade refinement network (CRN). In CRN, we designed a new cascade attention module (CAM) that uses multiple subnetworks and attention modules to aggregate the object features from different stages and progressively refine region proposals. CasA can be integrated into various two-stage 3-D detectors and improve their performance. Extensive experiments on the KITTI and Waymo datasets with various baseline detectors demonstrate the universality and superiority of our CasA. In particular, based on one variant of Voxel-region-based convolutional neural network (RCNN), we achieve the state-of-the-art results on the KITTI dataset. On the KITTI online 3-D object detection leaderboard, we achieve a high detection performance of 83.06%, 47.09%, and 73.47% average precision (AP) in the moderate car, pedestrian, and cyclist classes, respectively. Code is available at https://github.com/hailanyi/CasA**

***Index Terms*— 3-D object detection, cascade network, deep learning, light detection and ranging (LiDAR) point clouds.**

I. Introduction

THREE-DIMENSIONAL detection is one of the key tasks in scene understanding. The light detection and ranging (LiDAR) technology provides accurate 3-D spatial information in the form of point clouds. In recent years, 3-D object detection from LiDAR point clouds has gained more and more attention due to its wide application in smart cities [1], urban planning [2], and autonomous vehicles [3]. Compared with the well-studied 2-D detection problem, 3-D object detection from LiDAR point clouds is challenging as the data collected by LiDAR typically exhibit sparse and irregular distribution.

The existing methods follow single-stage or two-stage 3-D detection frameworks. The single-stage methods directly perform object detection using encoded features of point clouds [4], [5], [6], while the two-stage methods follow the region-based convolutional neural network (RCNN) framework presented in [7]. The latter approaches generate a set of candidate bounding boxes and then further classify and refine each candidate box. Many recent studies adopt the two-stage framework because of its higher accuracy. Multiview 3D network (MV3D) [8] generates object proposals from 2-D multiview feature maps, and then refines proposals using 2-D features. Point RCNN [9] and Voxel-RCNN [10] generate and refine region proposals using point and voxel features, respectively. These two-stage approaches usually refine region proposals using a single subnetwork. However, LiDAR scanning in 3-D space is uneven. There exists a huge distribution gap between nearby and distant objects. The single subnetwork has limited ability to learn objects under such diverse conditions, hindering the detection performance.

Some 2-D object detectors address a similar problem by integrating multiple separate subnetworks. For example, the multiregion detector [11] proposes an iterative bounding box regression that uses the same RCNN model several times to refine detection bounding boxes. Cascade RCNN [12], [13] cascades a series of RCNN heads and uses the output of one stage to train the next. In 3-D space, as a pioneering work, Meng *et al.* [14] presented a weakly supervised detector that directly applies the cascade framework for point clouds. Nevertheless, so far, the multistage framework for fully supervised 3-D object detection is still underexplored. Besides, directly applying the aforementioned cascade structures to 3-D object detection using a separate subnetwork only enhances/refines the object from each stage to a limited extent and often fails to achieve performance gain.

This article proposes **CasA**, a new **Cas**cade **A**ttention-based multistage framework, which performs 3-D object detection from vehicle-borne LiDAR point clouds. CasA consists of a region proposal network (RPN) and a cascade refinement

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network (CRN). The RPN uses a 3-D backbone network to encode voxels into 3-D feature volumes. Then a 2-D detection head is adopted to generate region proposals. Unlike most state-of-the-art two-stage 3-D detectors that refine region proposals using a single subnetwork, our CRN progressively refines and complements predictions from a series of subnetworks into high-quality predictions. Besides, different from the conventional cascade structures that use the output of one stage to train the next, we designed a new cascade attention module (CAM) to aggregate features from different stages for comprehensive region proposal refinement. In addition, CasA integrates a part-aided scoring that considers object completeness of parts as structure-aware single-stage 3-D object detection [structure-aware single-stage detector (SA-SSD)] [15] to better estimate proposal confidence. Compared with the recently released high-performance detectors [16], [17], our method uses only LiDAR points, while the above two methods use both LiDAR points and images. Moreover, the most advantage of our method is the expandability. Our cascade attention design can be extended to any two-stage 3-D detectors and greatly improves their detection performance. The experimental results demonstrated CasA's consistent performance improvement over multiple baseline detectors. For example, on the widely used KITTI [18] validation set (moderate car class), CasA improves PV-RCNN [19], Voxel-RCNN [10], and channel-wise transformer 3D detector (CT3D) [20] by 2.27%, 2.06%, and 0.57% average precision (AP) (R11), respectively. On the KITTI test set (moderate car class), CasA improves the Voxel-RCNN by 1.44% and achieves the state-of-the-art of 83.06% AP (R40). We believe that this effective design can be of interest to many 3-D downstream tasks.

Our contributions are summarized as follows.

1) We propose a new cascade framework, CasA, for 3-D object detection from LiDAR point clouds, which progressively refines and complements predictions through multiple subnetworks to obtain high-quality predictions. CasA can significantly improve the performance of various state-of-the-art 3-D object detectors.
2) We propose a CAM to aggregate object features from different stages. CAM comprehensively considers the quality of proposals from all the previous stages, significantly boosting the accuracy of proposal refinement.

II. Related Work

A. 3-D Object Detection From LiDAR Point Clouds

Most recent 3-D detectors follow a single-stage or two-stage framework. The single-stage methods directly perform object detection using encoded features of point clouds. Sparsely embedded convolutional detector (SECOND) [4], structure-information enhanced voxel network (SIEV-Net) [3], SA-SSD [15], and self-ensembling single-stage detector (SE-SSD) [21] first convert the point cloud into voxels, then abstract voxel features using the 3-D backbone network, and finally predict object confidences and boxes from the encoded features. 3DSSD [5] extracts point features by set abstraction layers to perform single-stage object detection. The two-stage methods usually first generate region proposals using an RPN, then extract the region features of the proposals using a region of interest (RoI) pooling method, and finally refine the proposal using the extracted features. Some previous methods generate and refine region proposals using multiview [8] or birds eye view (BEV) [22] features. PointRCNN [9] generates and refines region proposals using point features. Multiple methods [19], [23] encode point clouds into 3-D feature volumes using 3-D sparse convolution, compress the features into BEV representation to generate object proposals, and then refine the object proposals. Recent methods applied transformers for 3-D object detection. Votr [24] uses a Voxel Transformer to encode point clouds, and CT3D [20] applies a channel-wise transformer to refine object proposals.

B. Multistage Network for Object Detection

The multistage (beyond two-stage) methods have been more widely explored and demonstrated effective in 2-D object detection. Iterative bounding box regression [25] uses the same R-CNN model several times to refine detection bounding boxes. Integral loss [26] integrates the results from multiple proposal refinement networks to perform object detection. Cascade R-CNN [12], [13] cascades several detection networks, taking the output of the former as the input to train the latter. In this way, it progressively obtains results with raising intersection over unions (IoUs) and finally achieves significant performance gain. IoU-Net [27] improves Cascade R-CNN by predicting the IoU between the detected object and the matched ground truth. Gong *et al.* [28] introduced an LSTM-based proposal refinement module that iteratively refines bounding box proposals, further improving the 2-D detection performance. In 3-D space, as a pioneering work, Meng *et al.* [14] proposed a weakly supervised object detector that directly applies a cascade framework for point clouds.

However, multistage methods for fully supervised 3-D object detection on point clouds are still underexplored. In this work, we propose CasA, an effective multistage 3-D object detection framework that refines and complements predictions from multiple subnetworks into high-quality detections.

C. Attention-Based Network

Recently, the attention-based approach has been widely applied in the field of computer vision. For image, Dosovitskiy *et al.* [29] proposed a Vision Transformer that directly applies the attention mechanism to sequences of image patches. Detection transformer (DETR) [30] views object detection as a direct set prediction problem and builds an end-to-end object detector based on a transformer. Zhu *et al.* [31] proposed a spectral–spatial attention network for hyperspectral image classification. Cascaded feature attention and channel attention network (CFCA-Net) [32] proposed an cascaded feature attention method for cloud detection from satellite images. For point cloud, Zhao *et al.* [33] leverages the attention mechanism to process point cloud data and proposes the point transformer architecture as a general backbone. P4Transformer [34] uses self-attention to capture the appearance and motion information in point cloud video effectively. Inspired by these works, this article adopts an attention-based

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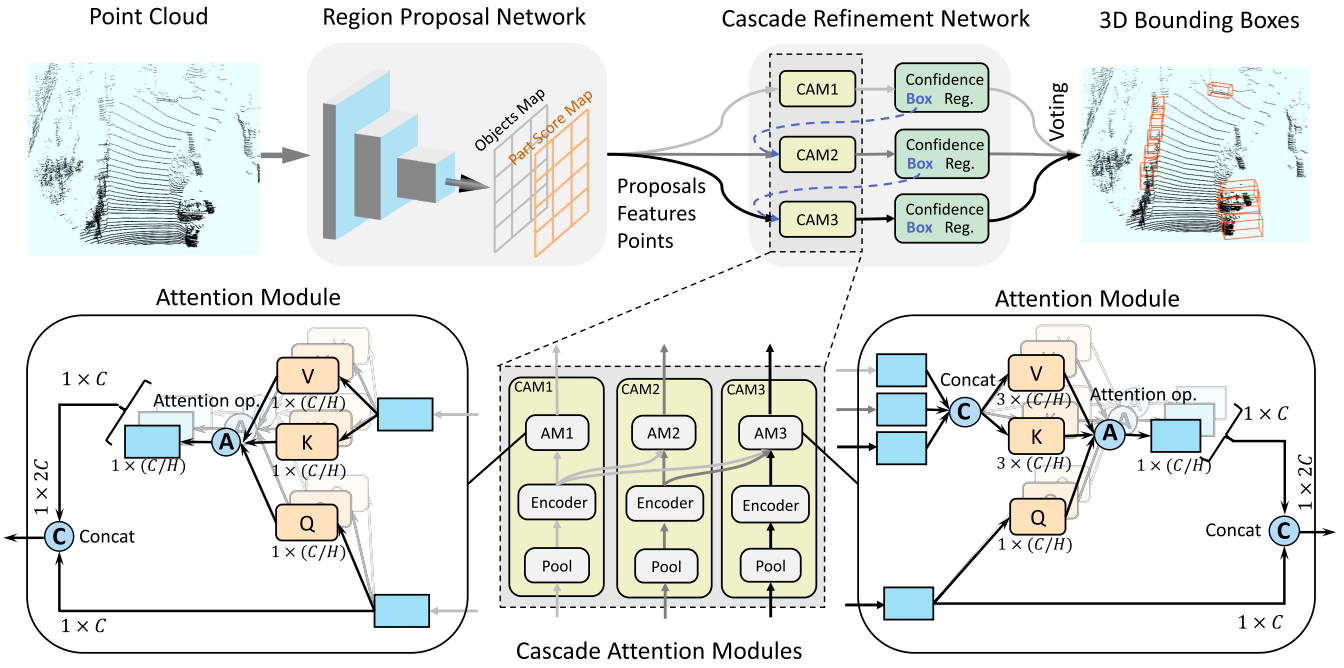


Fig. 1. Our CasA architecture. CasA generates proposals of 3-D bounding boxes using a voxel-based RPN, which consists of a 3-D backbone and a 2-D detection head. The proposals are progressively refined by a CRN, in which the features from different stages are aggregated by CAMs.

method to aggregate features from multiple subnetworks to achieve more accurate 3-D object detection.

III. Method

CasA is a multistage detection framework and can be integrated into various two-stage 3-D detectors. Current multistage methods and cascade structures [12] use a series of separate subnetworks to refine object proposals. Generally, these methods can learn object features under various difficult conditions. However, in these separate subnetworks, a later stage has limited ability to measure proposal quality in all the previous stages. This hinders the effectiveness of proposal refinement in the 3-D scene.

Our idea is to aggregate the features from all the stages in a cascade attention manner. As illustrated in Fig. 1, CasA consists of an RPN and a CRN. The RPN first uses a 3-D backbone network and a 2-D detection head to generate region proposals. The CRN consists of multiple subnetworks that progressively refine the proposals. In this CRN, we develop a new cascade attention scheme, which aggregates the proposal features from different stages for a more comprehensive bounding box prediction. We elaborate on this cascade attention design first.

A. Cascade Attention for Proposal Refinement

1) Vanilla Cascade Structure: The cascade detection framework is well-studied for 2-D images. Cascade R-CNN [12] uses a vanilla cascade structure, which uses a series of separate subnetworks with raising IoU thresholds to refine region proposals. Such a vanilla cascade structure consists of N_r refiners. The jth refiner takes a region proposal B^{j-1} from the previous stage as input and extracts the object feature F^j with a feature extractor $\phi(\cdot)$. Then, with F^j, a confidence prediction branch $\mathcal{S}(\cdot)$ and a box regression branch $\mathcal{R}(\cdot)$ output a new object confidence C^j and box B^j, respectively. This iterative refinement can be formulated as

$$F^j = \phi^j(B^{j-1}), \quad C^j = \mathcal{S}^j(F^j), \quad B^j = \mathcal{R}^j(F^j) \tag{1}$$

where $j = 1, 2, \ldots, N_r$. Such a design has been shown effective in 2-D object detection.

TABLE I

Results on the KITTI Validation Set (Moderate Car Class) by Applying a Vanilla Cascade Structure to Voxel-RCNN [10]. It Is Evaluated by AP (%) Under 40 Recall Thresholds

Detector	Test stages		
	1	2	3
Voxel-RCNN	85.27	-	-
+Vanilla Cascade	82.96	84.88	84.82

However, directly applying this vanilla cascade structure in 3-D does not bring desirable improvement. For example, on the KITTI [18] validation set, using the state-of-the-art Voxel-RCNN [10] detector with a vanilla cascade structure, the detection performance (on the moderate car class), as shown in Table I, does not improve.

The reasons for undesirable performance improvement are as follows.

1) Ignore Distant Objects: In multistage approaches, the later stages tend to be overfitting due to the lack of negative training samples. Two-dimensional approaches establish rising IoU thresholds to resample balanced samples. However, in 3-D point clouds, such resampling leads to unbalanced training between nearby and distant objects, because the point clouds are typically nonuniformly distributed. The nearby objects with dense points can produce high-quality proposals selected as positive samples, while distant objects tend to be negatives.

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Under such unbalanced training, a later stage predicts accurate nearby objects while ignoring distant objects. To tackle this problem, we increase more object appearances from all the previous stages, which ensures a later stage still has enough evidence to recover the ignored distant objects.

2) Error Propagation Problem: Three-dimensional detection is more challenging due to the need for object height and nonaxis aligned angle estimation. Small errors can propagate along with the downstream multistage framework, leading to detection failure. To address this, we build more connections between stages, and the errors from a single stage can be fixed by other stages in a complementary manner.

In short, our idea is to establish effective connections between these refiners to compose effective refinement. To this end, we designed a new feature extractor, named CAM, to aggregate object features across different stages.

2) Feature Aggregation Through Cascade Attention: As analyzed before, we aggregate the features across stages to increase object appearance to more accurately and robustly detect distant and hard objects.

Given a region proposal B^{j-1}, the majority of the existing detectors [19] apply a region pooling module $\mathcal{P}(\cdot)$ and a feature encoding module $\mathcal{E}(\cdot)$ to extract the proposal features $\hat{F}^j \in \mathbb{R}^{1\times C}$ for box regression and confidence prediction, where C is the feature dimension. In a cascade structure, however, such a strategy can only capture the proposal feature of the current stage while ignoring previous stages. In contrast, we further aggregate the features across stages. A naive method is directly concatenating the features from different stages. However, it is hard to learn the feature importance between stages, and it brings marginal performance improvement (see Table II). Inspired by the recent attention methods [24], we develop an attention-based operation to perform proposal features' aggregation from different stages.

For each encoded feature $\hat{F}^j$, we first concatenate a stage embedding P^j which is calculated by the position embedding in [35]. $\hat{F}^j = [\hat{F}^j, P^j]$. At the jth refinement stage, we collect the encoded features from all the previous stages and current stage $\mathbf{F}^j = [\hat{F}^0, \hat{F}^1, \ldots, \hat{F}^j]$. Then we have $\mathbf{Q}^j = \hat{F}^j\mathbf{W}_q^j$, $\mathbf{K}^j = \mathbf{F}^j\mathbf{W}_k^j$, and $\mathbf{V}^j = \mathbf{F}^j\mathbf{W}_v^j$, where $\mathbf{W}_q^j, \mathbf{W}_k^j$ and $\mathbf{W}_v^j$ are the linear projections. $\mathbf{Q}^j$, $\mathbf{K}^j$, and $\mathbf{V}^j$ are the query, key, and value embeddings, respectively. To enhance the representational ability, we also adopt multihead design. The embeddings from the ith head are denoted as $\mathbf{Q}_i^j$, $\mathbf{K}_i^j$ and $\mathbf{V}_i^j$. The attention value of a single head is calculated by

$$\hat{\mathbf{F}}_i^j = \text{softmax}\left(\frac{\mathbf{Q}_i^j\left(\mathbf{K}_i^j\right)^T}{\sqrt{C'}}\right)\mathbf{V}_i^j \tag{2}$$

where C' is the feature dimension in multihead attention Intuitively, the features from the current stage contribute more to proposal refinement. Hence, we also concatenate features $\hat{F}^j$ with the H multihead attention features to formulate the feature vector F^j which is used for box regression and confidence prediction

$$F^j = \text{Concat}\left(\hat{F}^j, \hat{\mathbf{F}}_1^j, \hat{\mathbf{F}}_2^j, \ldots, \hat{\mathbf{F}}_H^j\right). \tag{3}$$

For the first refinement stage, our module actually performs a self-attention operation. For other stages, we perform cross-attention operations that aggregate features from different stages. By adopting such a cascade attention design, our CasA can better estimate proposal quality in all the stages, which helps improve the proposal refinement accuracy.

3) Box Regression and Part-Aided Scoring: To perform a box regression, we follow [10], [19], which regress the box size, location, and orientation residuals relative to the input 3-D proposal. We also designed a part-aided score α^j to enhance confidence prediction (see Fig. 2). This is inspired by part-sensitive warping [15] that averages the object scores from a part score map. Such a design can help improve the confidence estimation, and we incorporate it into our CasA.

In our pipeline, we compute the part-aided score from both local (part-based) and global views. Specifically, α^j is computed by the confidence prediction branch $\mathcal{S}(\cdot)$ and the part-sensitive warping $\mathcal{W}(\cdot)$ as

$$\alpha^j = \mathcal{S}^j(F^j) + \mathcal{W}(B^j, \mathbf{X}) \tag{4}$$

where $\mathbf{X}$ is the part score map predicted by RPN. With this modification, this new part-aided scoring helps the detector more accurately estimate object confidence in each stage.

In training, similar to Cascade R-CNN [12], we set the 3-D IoU thresholds $u = \{u^1, u^2, \ldots, u^{N_r}\}$ to define the negatives and positives at different refinement stages. In testing,

TABLE II

ABLATION RESULTS ON THE KITTI VALIDATION SET USING DIFFERENT FEATURE AGGREGATION MODULES IN CASA + V

Aggregation Module	Car 3D (R40)(%)		
	Easy	Moderate	Hard
Concatenation	92.79	85.32	83.17
GRU	92.93	86.01	83.64
Self-attention	93.15	86.22	83.72
Cross-attention	93.01	86.29	83.85
Self-attention & Cross-attention	**93.21**	**86.37**	**83.93**

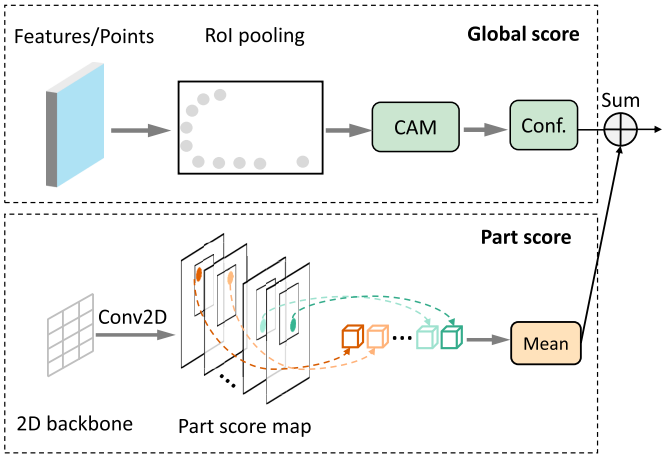


Fig. 2. Part-aided scoring. The object confidence is calculated by the sum of part scores from BEV map and global score from confidence branch.

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we average the boxes and scores from all the refinement stages to generate final detection results.

4) Boxes Voting: As analyzed in Section III-A1, 3-D detection is more challenging due to the need for object height and nonaxis aligned angle estimation. The errors tend to propagate along with the downstream multistage framework. To further address this problem, during testing, we propose boxes' voting to build more connections between stages. This is motivated by an intuition that each stage outputs both weak and strong predictions which can be ensembled together to generate more accurate predictions. Bearing this in mind, we explore methods to merge the boxes from all the refiners. A simple method is to directly perform nonmaximum suppression (NMS) on all the boxes, and it assembles the result by selecting the boxes with the highest confidence. However, it ignores lots of boxes with low confidence, which have the potential to recover missed objects. To address this, we adopt a weighted boxes' voting that directly averages the detection confidence and merges the boxes weighted by detection confidence as

$$C = \frac{1}{N_r} \sum_j C^j \tag{5}$$

$$B = \frac{1}{\sum_j C^j} \sum_j C^j \cdot B^j \tag{6}$$

where C and B are the merged confidence and box, respectively. After boxes' voting, we obtain a set of refined high-quality boxes. Nevertheless, there are still lots of redundant boxes as each object has many refined proposals. To remove redundant boxes, we finally perform an NMS on the voted results to produce detection outputs. By adopting the voting mechanism, various predictions (with less confidence, and from different perspectives/scales) generated by different refiners can compose, in a complementary manner, into more accurate/reliable final predictions.

B. Backbone Network

Many recent pipelines [4], [19] use a 3-D sparse convolution as backbone networks for accuracy and efficiency, and we also adopt this setting. We first split the raw points $\mathbf{P}$ into small voxels. For each voxel, we calculate the raw features using the mean of raw features of all the inside points. We adopt the 3-D sparse convolution $\mathcal{S}(\cdot)$ to encode 3-D point clouds into feature volumes. Here, $\mathcal{S}(\cdot)$ consists of a series of $3 \times 3 \times 3$ 3-D sparse convolution kernels, which downsample the spatial features to $1\times$, $2\times$, $4\times$, and eventually an $8\times$ downsampled tensor. The 3-D features in the last layer are compressed into BEV features along the height dimension for object proposal generation.

C. Region Proposal Network

We follow the recent works [10], [19] that generate object proposals by applying a series of 2-D convolution on BEV feature maps and generate the object proposals from BEV maps. Specifically, we first predefined N_p object templates called anchors on the last layer of BEV maps. We generate object proposals by classifying the anchors and regressing the residuals of object size, location, and orientation angle relative to ground-truth boxes. Similar to [10], [19], we assign the ground-truth bounding boxes to anchors by an IoU-based matching. For the ith anchor, we denote the score predict, score target, residual predict, and residual target as α_i, $\hat{\alpha}_i$, δ_i, and $\hat{\delta}_i^j$ respectively. The loss of proposal network is defined as

$$\mathcal{L}_{\text{RPN}} = \frac{1}{N_p} \left[\sum_i \mathcal{L}_{\text{score}}(\alpha_i, \hat{\alpha}_i) + \mathcal{I}(\text{IoU}_i > u) \sum_i \mathcal{L}_{\text{reg}}(\delta_i, \hat{\delta}_i) \right] \tag{7}$$

where $\mathcal{I}(\text{IoU}_i > u)$ indicates that only object proposals with $\text{IoU}_i > u$ produce the regression loss, and $\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{score}}$ are the smooth L1 and binary cross-entropy loss, respectively.

D. Overall Training Loss

Our CasA can be trained end-to-end by an RPN loss L_{RPN} and a CRN loss L_{CRN}. We combine the two losses with equal weights as $L = L_{\text{RPN}} + L_{\text{CRN}}$. The RPN loss is defined in Section III-C. The CRN loss is the summation of multiple refinement losses in multiple stages. In each refinement stage, we adopt the box regression loss $\mathcal{L}_{\text{reg}}$ and score loss $\mathcal{L}_{\text{score}}$ like [10], [19]. For the ith proposal at the jth refinement stage, we denote the score predict, score target, residual predict, and residual target as α_i^j, $\hat{\alpha}_i^j$, δ_i^j, and $\hat{\delta}_i^j$ respectively. The loss of CRN is defined as

$$\mathcal{L}_{\text{CRN}} = \frac{1}{N_b} \left[\sum_i \sum_j \mathcal{L}_{\text{score}}\left(\alpha_i^j, \hat{\alpha}_i^j\right) + \mathcal{I}\left(\text{IoU}_i^j > u^j\right) \sum_i \sum_j \mathcal{L}_{\text{reg}}\left(\delta_i^j, \hat{\delta}_i^j\right) \right] \tag{8}$$

where $\mathcal{I}(\text{IoU}_i^j > u^j)$ indicates that only object proposals with $\text{IoU}_i^j > u^j$ produce the regression loss.

IV. Experiments

A. Datasets and Evaluation Metrics

1) KITTI Dataset: The KITTI dataset includes 7481 and 7518 LiDAR frames for training and testing, respectively. We follow recent works [10] that divide the training data into a training split of 3712 frames and a validation split of 3769 frames. The primary official evaluation metric is a 3-D AP under 40 recall thresholds (R40). We also follow previous works [10], [19] and report the results on the validation set using a 3-D AP under 11 recall thresholds. The IoU thresholds in this metric are 0.7, 0.5, and 0.5 for cars, pedestrians, and cyclists, respectively.

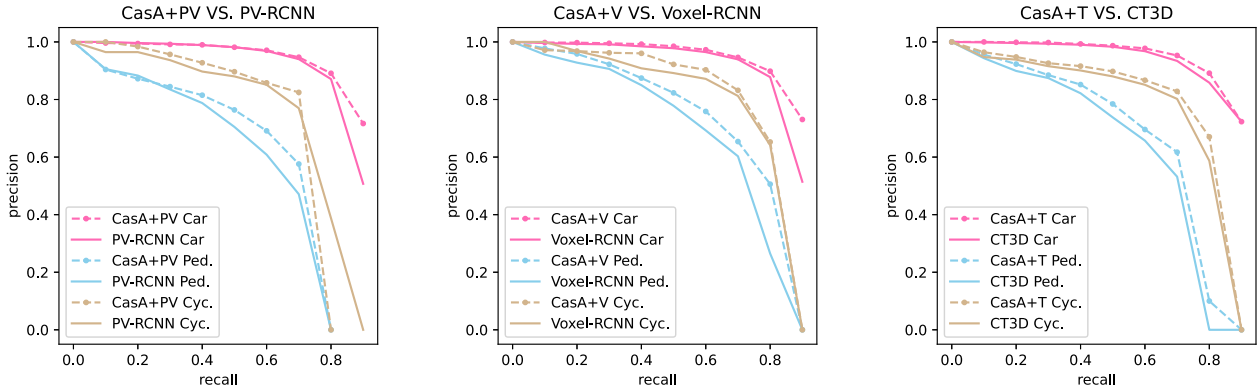
2) Waymo Open Dataset: The Waymo open dataset [36] contains 798 and 202 sequences with 158 361 and 40 077 LiDAR frames for training and validation, respectively. The official 3-D detection evaluation metrics are mean AP (mAP) (L1), mAP (L2), mAPH (L1), and mAPH (L2), where L1 and L2 denote the detection difficulty level. The mAPH metric takes into account object heading accuracy. The IoU thresholds

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TABLE III

THREE-DIMENSIONAL DETECTION RESULTS ON THE KITTI VALIDATION SET USING PV-RCNN, VOXEL-RCNN, AND CT3D AS BASE DETECTORS. THE PED. AND CYC. DETECTION RESULTS OF VOXEL-RCNN ARE REPRODUCED BY THEIR OPEN-SOURCE CODE. R40 AND R11 DENOTE AP UNDER 40 AND 11 RECALL THRESHOLDS, RESPECTIVELY. THE IMPROVED RESULTS ARE IN BOLD

Methods	Car 3D (R40)(%)			Ped. 3D (R40)(%)			Cyc. 3D (R40)(%)			Car 3D (R11)(%)		
	Easy	Mod.	Hard	Easy	Mod.	Hard	Easy	Mod.	Hard	Easy	Mod.	Hard
PV-RCNN [19]	92.57	84.83	82.69	64.26	56.67	51.91	88.88	71.95	66.78	-	83.90	-
CasA+PV	**92.73**	**85.89**	**83.57**	**68.90**	**59.86**	**53.66**	**90.95**	70.69	65.82	89.46	**86.17**	79.10
Voxel-RCNN [10]	92.38	85.29	82.86	68.22	60.97	55.63	91.28	72.54	68.46	89.41	84.52	78.93
CasA+V	**93.21**	**86.37**	**83.93**	**73.95**	**66.62**	**59.97**	**92.78**	**73.94**	**69.37**	**89.88**	**86.58**	**79.38**
CT3D [20]	92.85	85.82	83.46	65.73	58.56	53.04	91.99	71.60	67.34	89.54	86.06	78.99
CasA+T	**93.38**	**86.42**	**84.04**	**68.81**	**62.59**	**57.47**	**92.81**	**72.63**	**68.32**	**90.11**	**86.63**	**79.49**



(a) (b) (c)

Fig. 3. Recall–precision curves of our CasA on the KITTI validation set (moderate split). (a) Comparison between our CasA + PV and PV-RCNN. (b) Comparison between our CasA + V and Voxel-RCNN. (c) Comparison between our CasA + T and CT3D.

in this metric are 0.7, 0.5, and 0.5 for vehicles, pedestrians, and cyclists, respectively.

B. Setup Details

To demonstrate the universality and superiority of our CasA, we conducted our experiments on three popular baseline detectors: PV-RCNN [19], Voxel-RCNN [10], and CT3D [20]. We denoted the newly implemented detectors as CasA + PV, CasA + V, and CasA + T, respectively. For each detector on each dataset, we train a *single model* for three classes. We adopted three refinement stages (the number of stages is discussed in Section IV-E). In training, we adopted IoU thresholds $u = 0.5, 0.55, 0.6$ for vehicles and $u = 0.4, 0.45, 0.5$ for pedestrians and cyclists. We adopted the feature channel of 256 in our CAM.

1) CasA + PV: Our CasA + PV is constructed from PV-RCNN [19]. Directly applying CasA to PV-RCNN is impractical, as the 2-D backbone and Voxel Set Abstraction in RoI pooling are computationally intensive. It needs more GPU memory than we have. To tackle this, we simplified the 2-D backbone and RoI pooling in our CasA + PV: 1) the keypoint weighting branch in PV-RCNN is removed; 2) we only kept the 3-D features from the last two layers and BEV features for RoI-grid pooling; and (3) we set the feature dimension to half of the original feature dimension in the 2-D backbone.

2) CasA + V: Our CasA + PV is constructed from Voxel-RCNN [10]. We directly applied CasA to Voxel-RCNN as detailed in Section III.

3) CasA + T: Our CasA + T is constructed from CT3D [20]. Directly applying CasA to CT3D is also computationally intensive. We made some simplifications: 1) we set the feature dimension to half of the original feature dimension in the 2-D backbone and 2) we set the number of transformer encoding layers from three to two.

4) Training and Testing Details on KITTI: All the detectors use the same standard detection range ([0, 70.4] m for the X-axis, $[-40, 40]$ m for the Y-axis, and $[-3, 1]$ m for the Z-axis). All the detectors are trained on two 3090 GPU cards with a batch size of four. We kept the training setup the same as the baseline detectors. We used the Adam optimizer with a learning rate of 0.01 for CasA + PV and CasA + V, and 0.001 for CasA + T. Both CasA + PV and CasA + V are trained for 80 epochs, while CasA + T is trained for 100 epochs (following CT3D [20]). During training, we set NMS IoU 0.8 to keep 128 proposals from RPN with 1:1 ratio of positive and negative samples. Regular data augmentation methods, such as flipping, rotation, scaling, and ground-truth sampling, are also adopted. During testing, we kept the top 100 proposals for cascade refinement, and finally use NMS IoU 0.1 to produce the final detection results.

5) Training and Testing Details on Waymo: All the detectors use the same standard detection range of $[-75.2, 75.2]$ m for the X- and Y-axes and $[-2, 4]$ m for the Z-axis. The detectors are trained on eight V100 GPU cards with a batch size of 16 for 30 epochs. The Adam optimizer with a learning rate of 0.01 is adopted for CasA + PV and CasA + V, and 0.001 for CasA + T. During training, we set the NMS IoU to be 0.8 to keep 128 proposals. Regular data augmentations are adopted (see the above paragraph). During testing, we kept the top 500 proposals, and use NMS IoU 0.3 to produce the final detection results.

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TABLE IV

THREE-DIMENSIONAL DETECTION RESULTS ON THE KITTI TEST SET. THE BEST METHODS ARE IN BOLD. ONLY PUBLISHED METHODS ARE REPORTED. OUR CASA + V OUTPERFORMS ALL THE METHODS ON ALL THE THREE CLASSES (CAR, PEDESTRIAN, AND CYCLIST)

Method	Car 3D (R40)(%)			Ped. 3D (R40)(%)			Cyc. 3D (R40)(%)		
	Easy	Mod.	Hard	Easy	Mod.	Hard	Easy	Mod.	Hard
LiDAR+RGB									
MV3D [8]	74.97	63.63	54.00	-	-	-	-	-	-
F-PointNet [37]	82.19	69.79	60.59	50.53	42.15	38.08	72.27	56.12	49.01
F-ConvNet [37]	87.36	76.39	66.69	52.16	43.38	38.8	81.98	65.07	56.54
UberATG-MMF [38]	88.40	77.43	70.22	-	-	-	-	-	-
EPNet [39]	89.81	79.28	74.59	52.79	44.38	41.29	-	-	-
3D-CVF [40]	89.20	80.05	73.11	-	-	-	-	-	-
CLOCs [41]	88.94	80.67	77.15	-	-	-	-	-	-
LiDAR									
PointRCNN [9]	86.96	75.64	70.70	47.98	39.37	36.01	74.96	58.82	52.53
3D Iou Loss [42]	86.16	75.64	70.70	-	-	-	-	-	-
STD [43]	87.95	79.71	75.09	53.29	42.47	38.35	78.69	61.59	55.30
HotSpotNet [44]	87.60	78.31	73.34	53.10	45.37	41.47	82.59	65.95	59.00
Part A^2 [23]	87.81	78.49	73.51	53.10	43.35	40.06	79.17	63.52	56.93
Point-GNN [45]	88.33	79.47	72.29	51.92	43.77	40.14	78.60	63.48	57.08
3DSSD [5]	88.36	79.57	74.55	50.64	43.09	39.65	82.48	64.10	56.90
SA-SSD [15]	88.75	79.79	74.16	-	-	-	-	-	-
PV-RCNN [19]	90.25	81.43	76.82	52.17	43.29	40.29	78.60	63.71	57.65
CIA-SSD [46]	89.59	80.28	72.87	-	-	-	-	-	-
CT3D [20]	87.83	81.77	77.16	-	-	-	-	-	-
VoTr [24]	89.90	82.09	79.14	-	-	-	-	-	-
Pyramid-PV [47]	88.39	82.08	77.49	-	-	-	-	-	-
SPG [48]	90.50	82.13	78.90	-	-	-	-	-	-
SE-SSD [21]	91.49	82.54	77.15	-	-	-	-	-	-
BtcDet [49]	90.64	82.86	78.09	47.80	41.63	39.30	82.81	68.68	61.81
Voxel-RCNN [10]	90.90	81.62	77.06	-	-	-	-	-	-
CasA+V	**91.58**	**83.06**	**80.08**	**54.04**	**47.09**	**44.56**	**87.91**	**73.47**	**66.17**

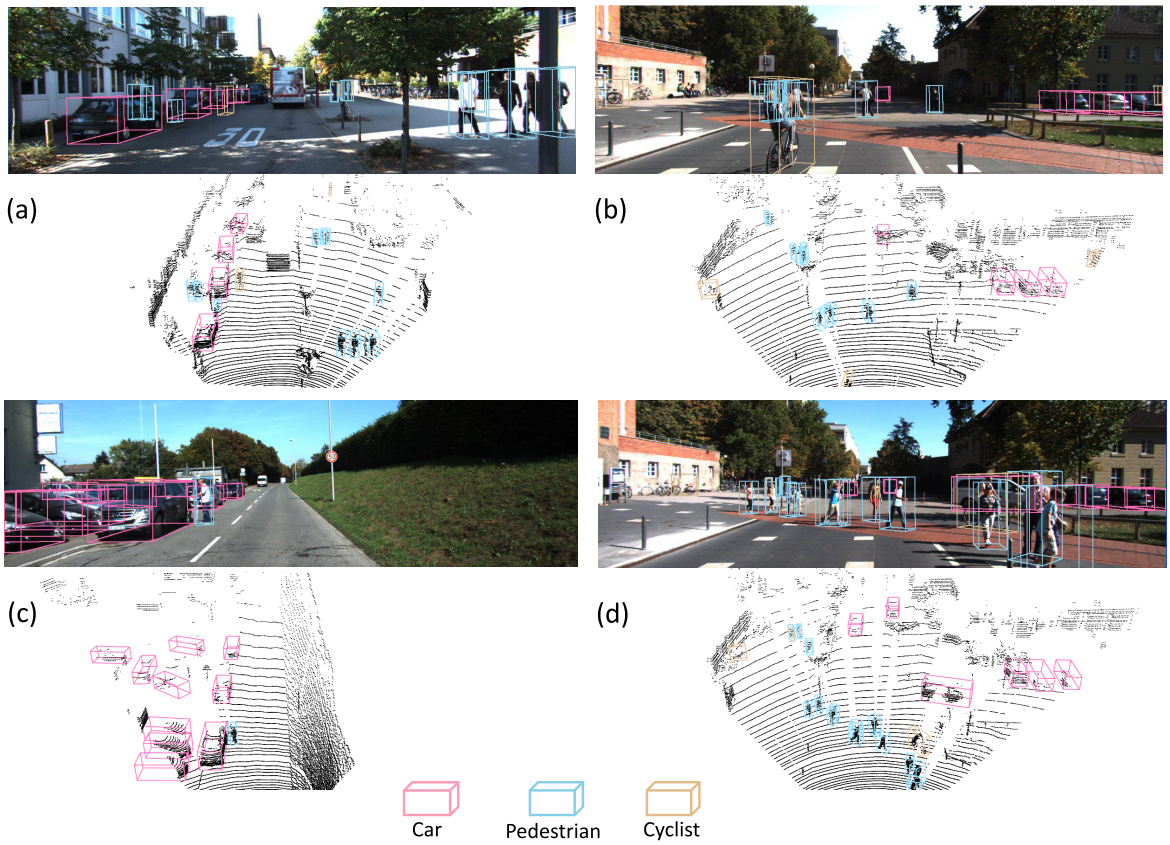


Fig. 4. Example qualitative results of CasA + V on the KITTI test set. We show our detected results of four different scenes in (a)–(d), in which car, pedestrian, and cyclist are shown in hotpink, skyblue, and tan, respectively.

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TABLE V

THREE-DIMENSIONAL DETECTION RESULTS ON THE WAYMO VALIDATION SET USING PV-RCNN, VOXEL-RCNN, AND CT3D AS BASE DETECTORS. †: REIMPLEMENTED RESULTS OF PV-RCNN REPORTED IN [50]. ‡: REIMPLEMENTED BY OURSELVES USING THEIR CODES. THE IMPROVED RESULTS ARE IN BOLD

Methods	Veh.(L1)(%)		Veh.(L2)(%)		Ped.(L1)(%)		Ped.(L2)(%)		Cyc.(L1)(%)		Cyc.(L2)(%)	
	mAP	mAPH	mAP	mAPH	mAP	mAPH	mAP	mAPH	mAP	mAPH	mAP	mAPH
†PV-RCNN [19]	77.51	76.89	68.98	68.41	75.01	65.65	66.04	57.61	67.81	66.35	65.39	63.98
CasA+PV	**78.55**	**78.06**	**69.67**	**69.23**	**77.22**	**70.60**	**68.06**	**62.08**	**68.19**	**66.76**	**65.73**	**64.33**
‡Voxel-RCNN [10]	77.43	76.71	68.73	68.24	76.37	68.21	67.92	60.40	68.74	67.56	66.46	65.35
CasA+V	**78.54**	**78.00**	**69.91**	**69.42**	**80.88**	**73.10**	**71.87**	**64.78**	**69.66**	**68.38**	**67.07**	**66.83**
‡CT3D [20]	77.75	77.22	68.92	68.44	75.38	68.08	66.17	59.60	69.28	67.88	66.84	65.48
CasA+T	**78.26**	**77.72**	**69.57**	**69.12**	**76.83**	**70.51**	**68.36**	**62.51**	**71.40**	**70.04**	**68.86**	**67.55**

TABLE VI

ABLATION RESULTS ON THE KITTI VALIDATION SET USING DIFFERENT CASCADE STAGES IN CASA + V

Cascade stages	Car 3D (R40)(%)			Run time (ms)
	Easy	Moderate	Hard	
1	92.84	85.49	83.04	64
2	93.06	85.96	83.58	77
3	93.21	**86.37**	**83.93**	86
4	**93.30**	86.27	83.86	101
5	93.18	86.24	83.87	114

TABLE VII

ABLATION RESULTS ON THE KITTI VALIDATION SET USING DIFFERENT NUMBERS OF ATTENTION HEADS IN CASA + V

Number of attention heads	Car 3D (R40)(%)		
	Easy	Moderate	Hard
1	92.89	86.11	81.77
4	**93.21**	**86.37**	**83.93**
8	93.07	86.27	83.74
16	93.04	86.17	83.64

C. Evaluation on the KITTI Dataset

1) Validation Set: We first conducted experiments on the KITTI validation set. The results are shown in Table III and Fig. 3. Our CasA outperforms the baseline Voxel-RCNN, PV-RCNN, and CT3D with 2.27%, 2.06%, and 0.57% AP (R11), respectively, in the moderate car class. Notably, our CasA improves the pedestrian detection performance by 3.19%, 5.65%, and 4.03%, respectively. The performance gains are mostly derived from the cascade attention design that aggregates proposal features from multiple stages, leading to a more effective and comprehensive object refinement. Note that the cyclist detection performance of CasA + PV slightly drops, mostly because we simplified the 2D backbone of the baseline detector.

2) Test Set: To further demonstrate the advance of our cascade attention design, we trained our CasA + V using all the training data of the KITTI training set. The results on the KITTI test set are summarized in Table IV. Our CasA + V outperforms all the published methods and achieves state-of-the-art performance. We note that our CasA + V outperforms all the previous methods in all the three classes (car, pedestrian, and cyclist). We show some qualitative results of CasA + V on the KITTI test set in Fig. 4.

D. Evaluation on the Waymo Open Dataset

The results on the Waymo validation set are shown in Table V. Our CasA improved all the baseline detectors on all the metrics by a large margin. Specifically, compared with the PV-RCNN [19], Voxel-RCNN [10], and CT3D [20], our CasA + PV, CasA + V, and CasA + T improve the vehicle detection mAPH by 0.82%, 1.18%, and 0.68%, respectively. Our cascade design significantly improves the pedestrian detection performance by 4.47%, 4.38%m and 2.91% mAPH, respectively. The results further demonstrate the effectiveness of our method.

TABLE VIII

ABLATION STUDY RESULTS OF PART-AIDED SCORING, CAM, AND BOXES' VOTING ON THE KITTI VALIDATION SET (MODERATE CAR CLASS)

Structure	Car 3D (R40)(%)		
	Stage 1	Stage 2	Stage 3
Voxel-RCNN	85.27	-	-
+Vanilla Cascade	82.96	84.88	84.82
+Cascade Attention	85.48	85.73	85.72
+Part-aided Scoring	85.67	85.89	85.85
+Boxes Voting	85.67	86.24	**86.37**

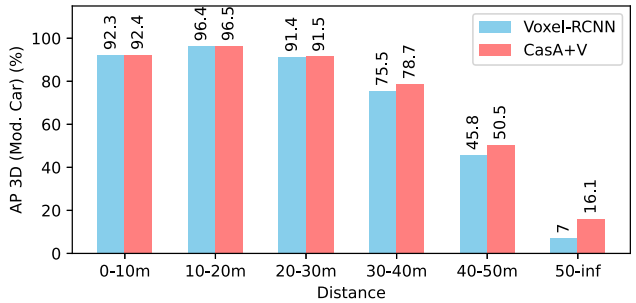


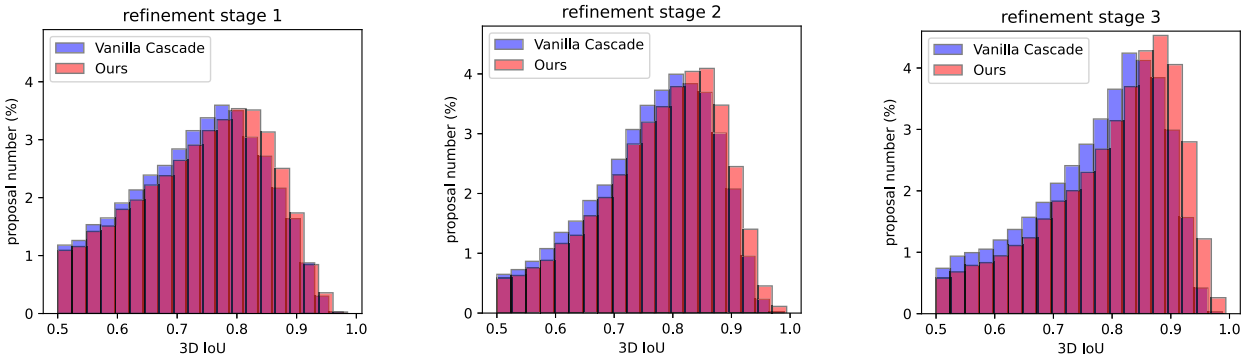
Fig. 5. Ablation results on the KITTI validation set for different distances. Compared with the Voxel-RCNN baseline, our CasA + V has a better detection performance of distant objects.

The results on the Waymo validation set are shown in Table V. Our CasA improved all the baseline detectors on all the metrics by a large margin. Specifically, compared with PV-RCNN [19], Voxel-RCNN [10], and CT3D [20], our CasA + PV, CasA + V, and CasA + T improve the vehicle detection mAPH by 0.82%, 1.18%, and 0.68%, respectively. Our cascade design significantly improves the pedestrian detection performance by 4.47%, 4.38%, and 2.91% mAPH, respectively. The results further demonstrate the effectiveness of our method.

E. Ablation Study

We conducted experiments on the KITTI validation set (moderate car class) and used CasA + V to examine the

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(a) (b) (c)

Fig. 6. Proposal quality in (a) refine stage 1, (b) refine stage 2, and (c) refine stage 3. With the increase in stages, the 3-D IoUs between the proposals and ground truths become closer to 1. Our cascade attention design also shows better IoUs compared with the vanilla cascade structure. It demonstrates the effectiveness of our design.

hyperparameters and each component/design of the proposed method.

1) Cascade Stages: We first tested the cascade stages. The results are shown in Table VI. CasA + V achieves the best performance on moderate and hard car class using three stages while achieving the best performance on easy car class using four stages. We observe that the detection performance of the three and four stages is close to each other. For computational efficiency, we adopt three stages. It can also achieve real-time performance with a running speed of 86 ms (single 3090 GPU).

2) Number of Attention Heads: In our CAM, we use multiple attention heads. To find the best head number, we conducted an ablation experiment. The results are shown in Table VII. Using four heads, our CasA + V achieves the best performance. Therefore, we adopt four heads in this article.

3) Effectiveness of CAM: To investigate the effects of CAM, we first constructed a baseline of the vanilla cascade structure, which uses Voxel-RCNN [10] as the base detector and integrates a simple cascade structure with raising IoU thresholds. The comparison results are shown in Table VIII. A simple cascade structure cannot bring performance improvement. By adding our proposed CAM, the result is further improved to 85.72%. Our CAM aggregates object features from different detection stages, producing more robust detections from scenes with sparse points. Especially for the distant objects (more than 40 m), our module improves the baseline by around 5% AP (see Fig. 5). We also tested alternative aggregation methods such as concatenation, GRU, only self-attention, and only cross-attention. The results are shown in Table II. The module using both self-attention and cross-attention achieves the best performance on easy (93.21%), moderate (86.37%), and hard (83.93%) car classes among the above aggregation modules. Therefore, our method uses the self-attention and cross-attention design as the aggregation strategy.

4) Effectiveness of Part-Aided Scoring: We also conducted an ablation study on part-aided scoring, and the results are also shown in Table VIII. By adding this design, our method achieves the results of 85.85%, which indicates the effectiveness of the part-aided scoring design.

5) Effectiveness of Boxes' Voting: As shown in Table VIII, the boxes' voting further improved the 3-D detection performance of moderate car class from 85.85% to 86.37%. The reason is that our method can now progressively boost and complement weak predictions from each cascade stage and thus generate more accurate predictions.

TABLE IX
RESULTS ON THE ONCE VALIDATION SET

Method (multi-class model)	AP 3D (%)		
	Vehicle	Pedestrian	Cyclist
Voxel-RCNN [10]	77.12	41.90	62.31
CasA+V	**78.34**	**50.34**	**70.01**

F. Proposal Quality in Cascade Attention Design

In cascade RCNN [12], the proposal quality becomes better and better with the increase in stages. This leads to a better performance in 2-D space. In 3-D space, here, we also demonstrate the same property of CasA. With the increase in stages, the 3-D IoUs between the proposals and ground truths become closer to 1. Besides, our design obtains higher IoUs than the vanilla cascade structure. It demonstrates the effectiveness of our cascade attention design. We show the results in Fig. 6.

G. Results on Dataset With Sparser LiDAR Scanning

The Waymo and KITTI datasets are collected by a 64-beam LiDAR sensor, and the acquired points are relatively dense. To validate our method on various resolutions, we also conducted an experiment on the ONCE dataset [51], which is collected by a 40-beam LiDAR sensor. The points of ONCE are also sparser than the Waymo and KITTI datasets. We used the metrics, the official training and validation data split provided by ONCE. The results of the ONCE validation set are shown in Table IX. Compared with the Voxel-RCNN baseline, our CasA + V has 1.2%, 8.4%, and 7.7% AP improvement on vehicle, pedestrian, and cyclist, respectively. We can observe from the results that our cascade attention design performs better than the Voxel-RCNN baseline on the low-resolution dataset. These results further demonstrate the effectiveness of our method.

V. CONCLUSION

This article presented a multistage 3-D object detector, CasA, which progressively refines region proposals by a

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cascade attention structure. CasA addressed the problems of ignoring distant objects and error propagation in multistage 3-D object detection by aggregating object features from multiple stages. CasA can significantly improve the performance of various state-of-the-art 3-D object detectors. We verified this design on two datasets with three different popular detectors. On the widely used KITTI validation set (moderate car class), CasA improves PV-RCNN, Voxel-RCNN, and CT3D by 2.27%, 2.06%, and 0.57% AP(R11), respectively. This fully demonstrated the effectiveness and generality of our method. We believe that this design can be of interest to many 3-D downstream tasks such as object tracking and motion prediction. With a cascade attention design, our CasA method achieves promising object detection performance on multiple datasets. Compared with previous approaches, the performance is significantly better in the case of three-stage refinement. However, there is a trade-off between computational cost and performance improvement. Future work will focus on developing approaches that can refine proposals more efficiently. Furthermore, pedestrians are more difficult to detect than other classes (see Tables IV, V, and IX) due to their small size and difficulty in distinguishing directions. Future work will also focus on detecting small objects more accurately.

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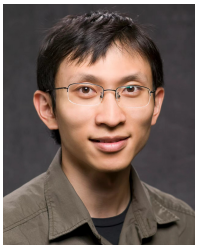
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